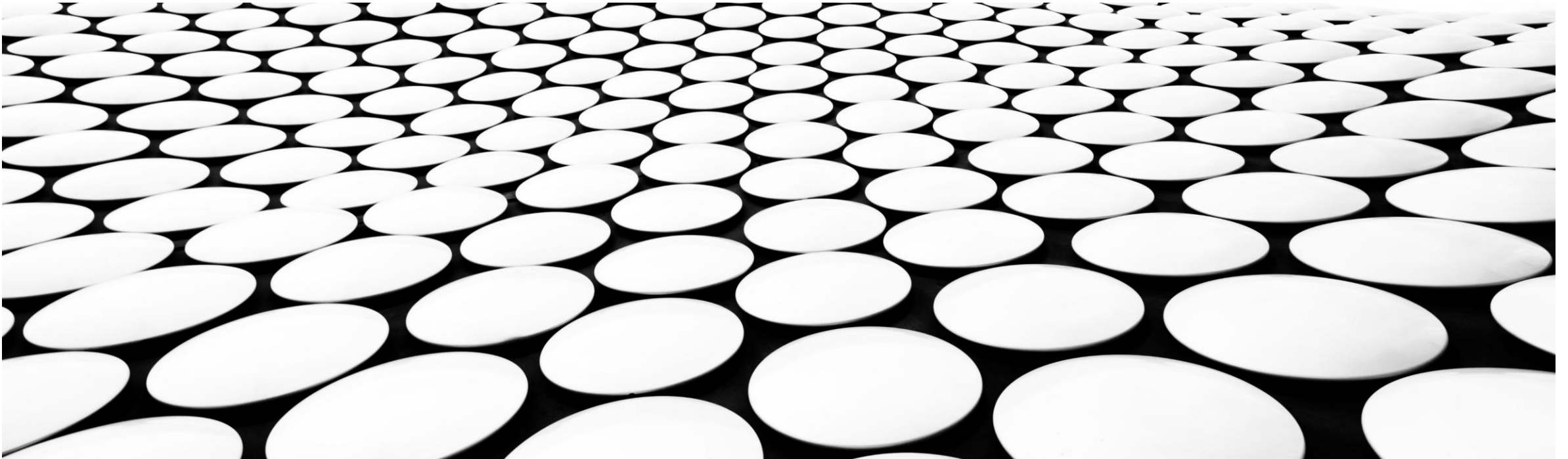


DATA CLEANING

Toward better quality data





DATA CLEANING

- Data wrangling
- Data quality
- Data cleaning
 - Duplicates
 - Invalid data
 - Outliers

Part 1

Part 2

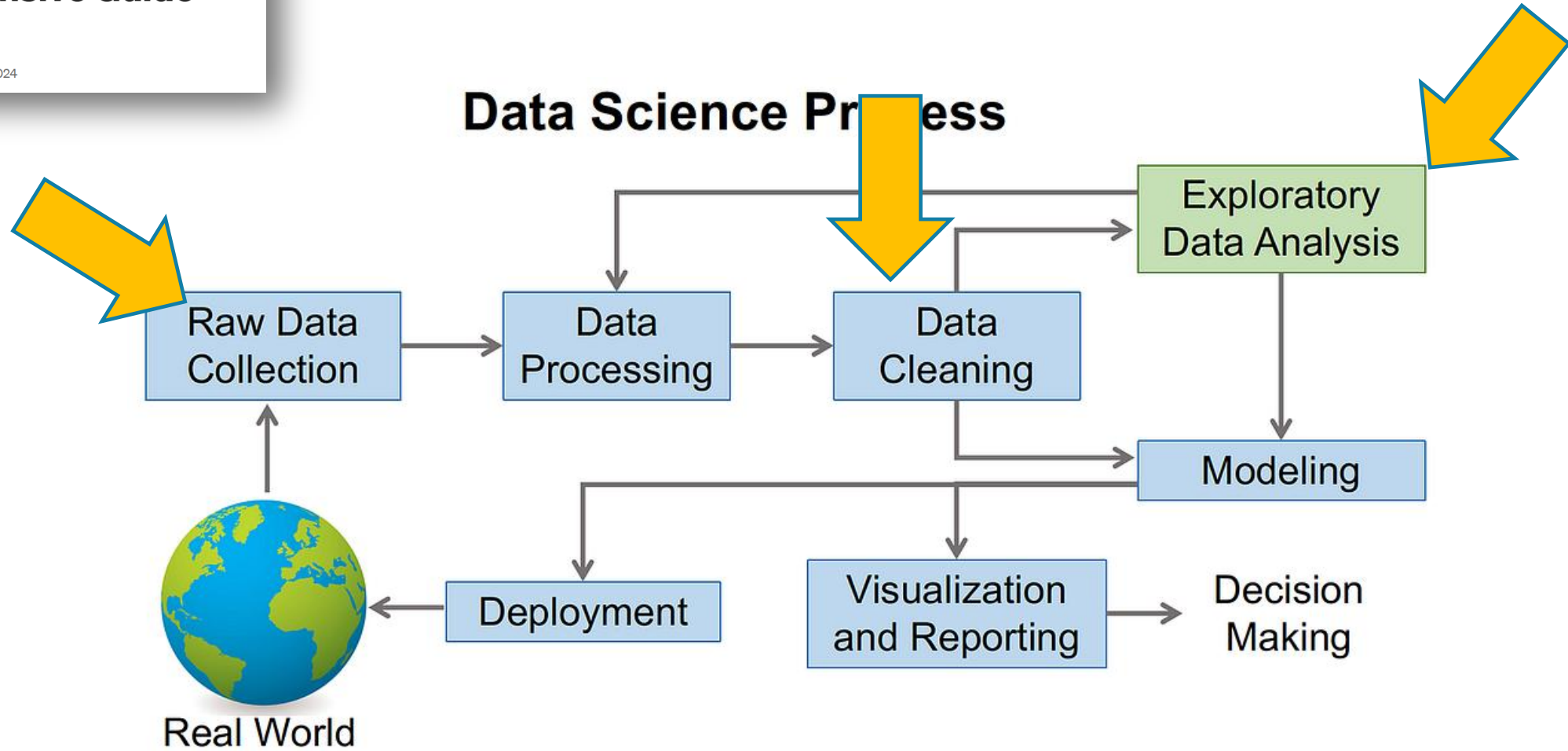
I'll present imputation methods
for missing data later (week 6)

DATA WRANGLING

Data Science Process: A Comprehensive Guide



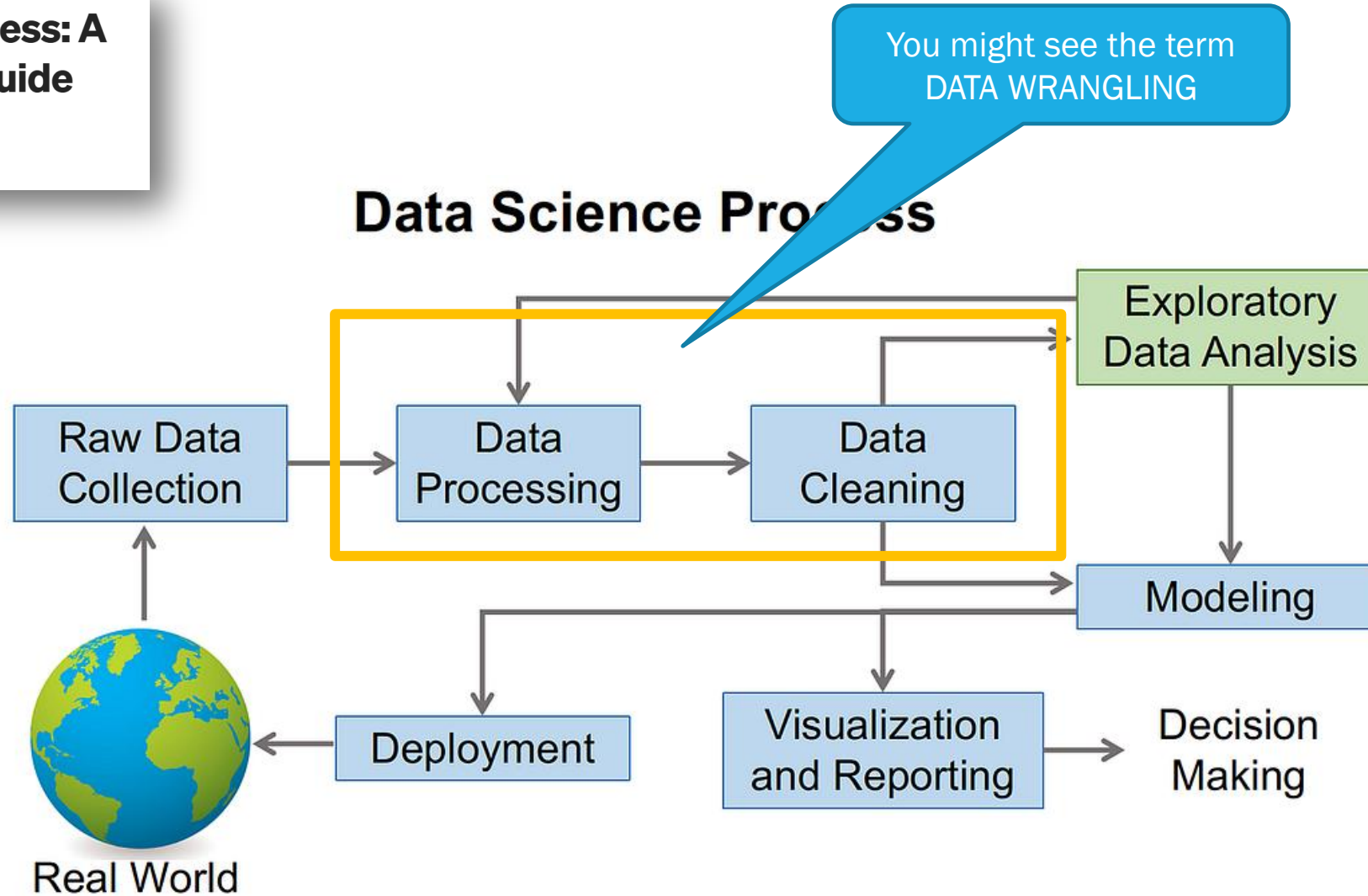
Abhijit · Follow
6 min read · Jan 15, 2024



Data Science Process: A Comprehensive Guide



Abhijit · Follow
6 min read · Jan 15, 2024



DATA WRANGLING: WHAT IT IS & WHY IT'S IMPORTANT



19 JAN 2021

Tim Stobierski | Contributors

Data Wrangling vs. Data Cleaning

Despite the terms being used interchangeably, data wrangling and data cleaning are two different processes.

It's important to make the distinction that data cleaning is a critical step in the data wrangling process to remove inaccurate and inconsistent data. Meanwhile, data-wrangling is the overall process of transforming raw data into a more usable form.

Data Wrangling vs. Data Cleaning

DATA CLEANING



Removing inaccurate
and inconsistent data

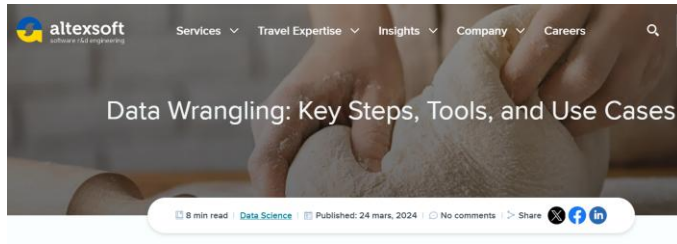
DATA WRANGLING

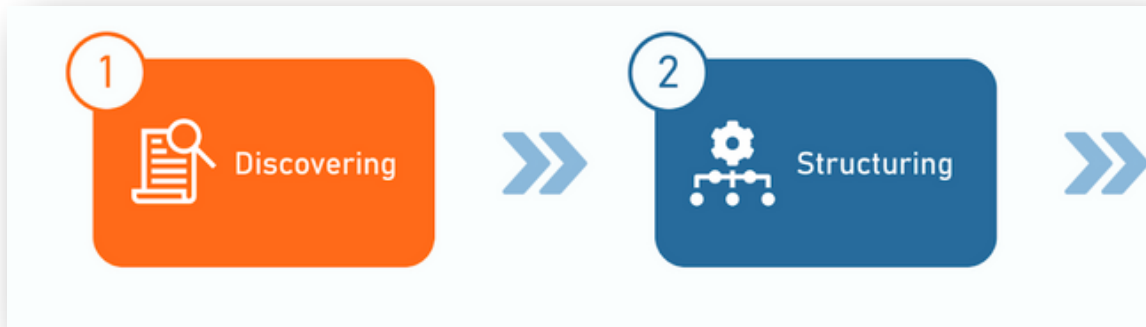


Transforming raw data
into a more usable form



Harvard
Business
School
Online



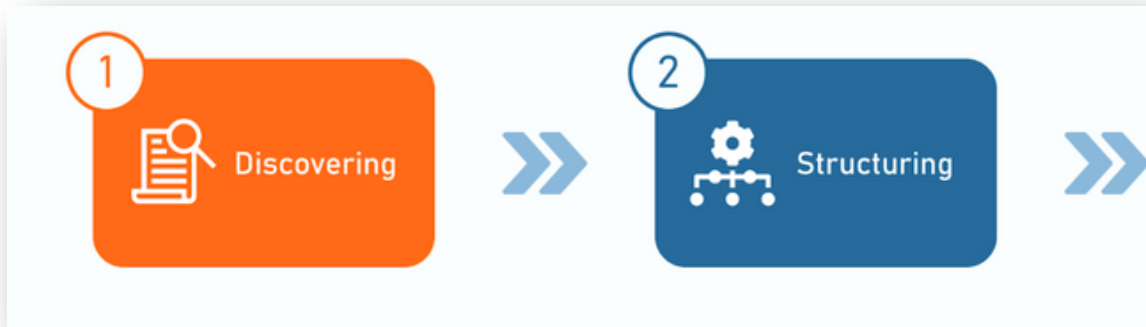


Data collection

- Sensor Data
- Data obtained through API
- Data extracted from Social Media

Transform into tabular data

Convert unstructured data to structured data



Example of API response

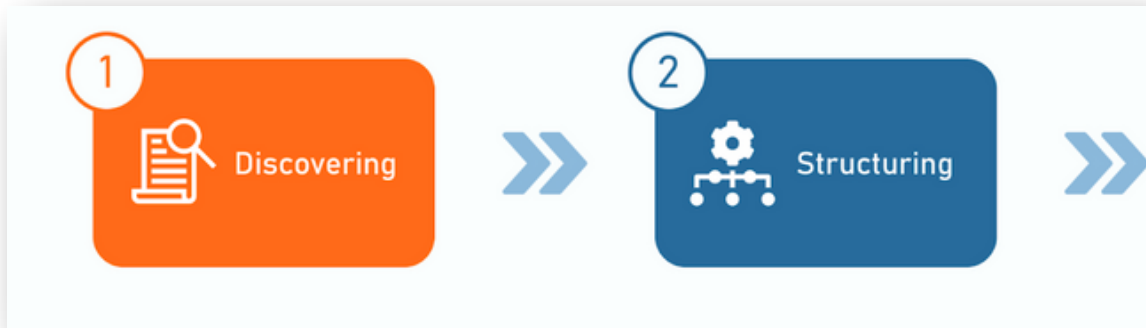
```
{
  "lat":33.44,
  "lon":-94.04,
  "timezone":"America/Chicago",
  "timezone_offset":-18000,
  "current":{
    "dt":1684929490,
    "sunrise":1684926645,
    "sunset":1684977332,
    "temp":292.55,
    "feels_like":292.87,
    "pressure":1014,
    "humidity":89,
    "dew_point":290.69,
    "uvi":0.16,
    "clouds":53,
    "visibility":10000,
    "wind_speed":3.13,
    "wind_deg":93,
    "wind_gust":6.71,
    "weather":[
      {
        "id":803,
        "main":"Clouds",
        "description":"broken clouds",
        "icon":"04d"
      }
    ]
  },
}
```

API call

```
https://api.openweathermap.org/data/3.0/onecall?lat=
{lat}&lon={lon}&exclude={part}&appid={API key}
```



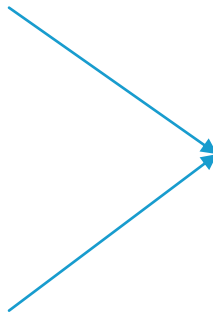
<https://openweathermap.org/api/one-call-3>



This is complex enough that we will hopefully have time to come back to this topic,,,

Dataset 1

Dataset 2



Gather and merge

Case Study
Energy Consumption



Sensors

APIs

Going back to example
from last week

Sensor Types Included

1. Smart electricity meter
2. Smart thermostat
3. Indoor temperature sensor
4. Occupancy / motion sensor

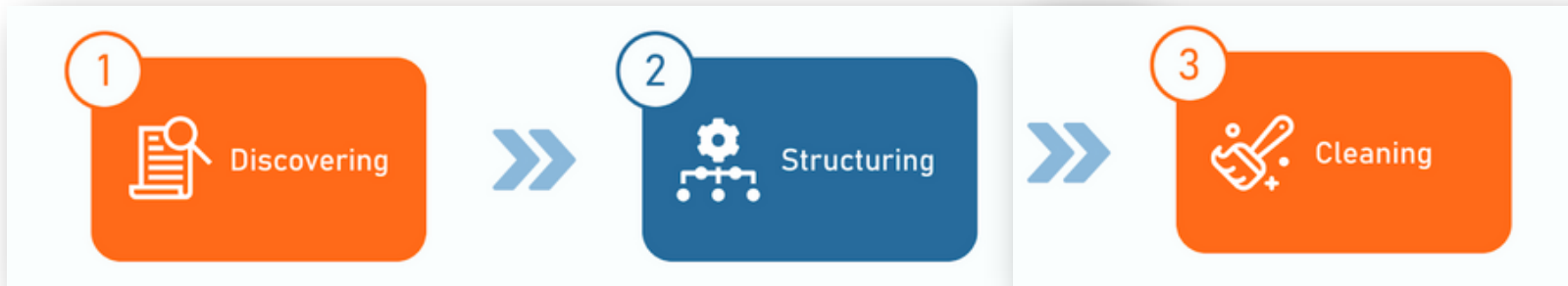
Merging on TimeStamp, but
with non-corresponding times.
Time series analysis (end of
semester)

Electricity Smart Meter (Whole-Home)

timestamp	meter_id	power_kW
2025-02-12 06:00:00	MTR_88421	0.8
2025-02-12 06:15:00	MTR_88421	1.1
2025-02-12 06:30:00	MTR_88421	1.6
2025-02-12 06:45:00	MTR_88421	2.3
2025-02-12 07:00:00	MTR_88421	3.0

Smart Thermostat

timestamp	thermostat_id	setpoint_C	indoor_temp_C	hvac_mode
2025-02-12 05:50:12	THM_33210	21.0	19.4	heating
2025-02-12 06:10:08	THM_33210	21.0	19.9	heating
2025-02-12 06:28:44	THM_33210	22.0	20.3	heating
2025-02-12 06:47:01	THM_33210	22.0	20.9	heating
2025-02-12 07:15:33	THM_33210	20.0	21.2	off



Data cleaning includes data profiling, which is the process of reviewing the content and quality of a dataset to detect potential issues or anomalies.

But... quality is more than that.

DATA QUALITY



The 6 data quality dimensions with examples

Data quality goes beyond what cleaning can do...

1. Completeness

How complete is your dataset?

2. Accuracy

Is the data an accurate representation of the element it describes?

3. Consistency

Is your data synchronized across your organization?

4. Validity

Does the data fit the defined range and definition?

5. Uniqueness

Are there any duplicates within your data?

6. Integrity

Can your data be traced and connected across the organization?

1. Completeness

How complete is your dataset?



SURVEY

Population targetted is the proper one ?

- University students versus Computer Science students

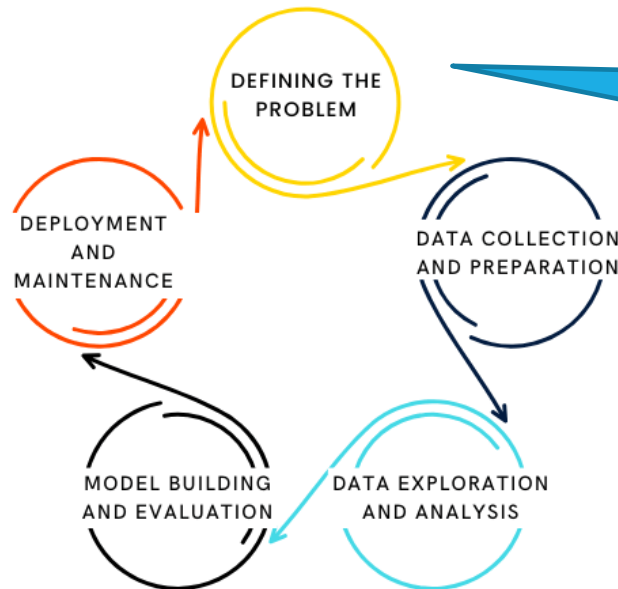
Attributes gathered sufficient ?

- Hours of sleep per week was not asked but we would like to find correlations with academic achievement.

SECONDARY DATA

Data does not contain required attributes.

What Are The 5 Steps in Data Science Lifecycle



Completeness is related to a good problem definition and good planning of the data collection

2. Accuracy

Is the data an accurate representation of the element it describes?



Does the information reflect reality?

- Price of food (where is it taken?)
- Age of employees (how is it entered)

This is related to trust of sources/users.

Cleaning can detect anomalies... but does not replace a trustable source!



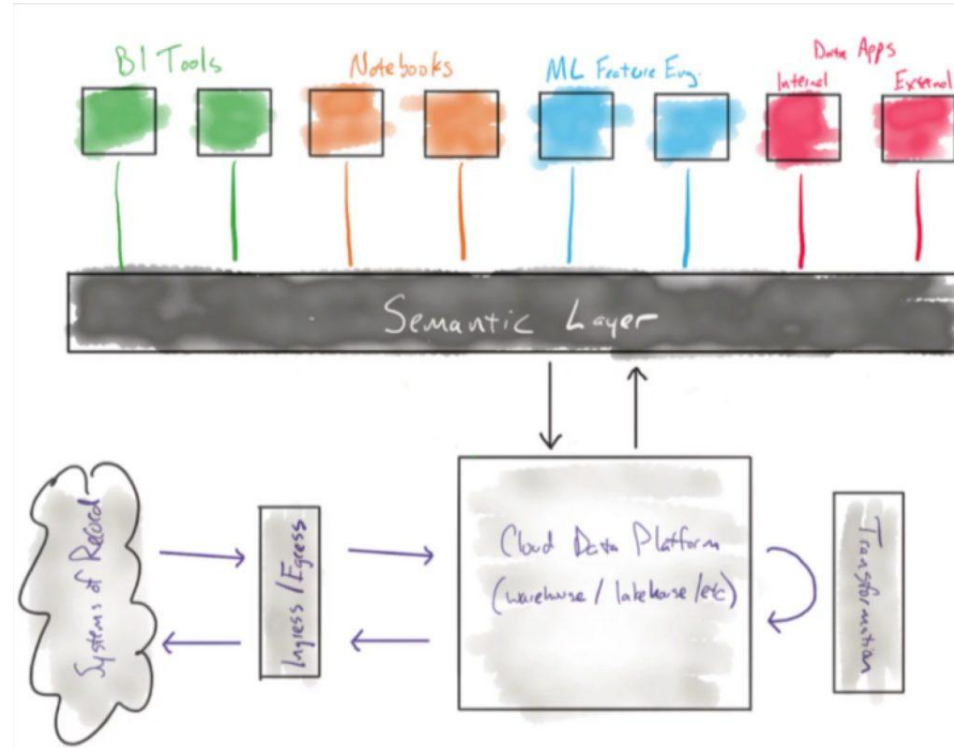
20\$/kg

3. Consistency

Is your data
synchronized across
your organization?



Unique dataset with shared views within the organization?
Data fields different from dataset to another?



Column names (surface form)
How-Much, Cost, Price, Amount

How much money is paid (meaning)

4. Validity

Does the data fit the defined range and definition?



Rules on attributes

- Salary cannot be negative

Rules on pairs of attributes

- Job title cannot be « Manager » and salary smaller than X

A LOT of cleaning is about these types of errors

5. Uniqueness

Are there any duplicates within your data?



Redundant data can impact results in descriptive and predictive analysis.

Removing duplicates is always mentioned in cleaning

But what about near-duplicates

6. Integrity

Can your data be traced and connected across the organization?

I let you explore this term!!

Sometimes defined as part of Data Governance...
Sometimes defined as encompassing Data Quality....

Whenever something changes and such change cannot be traced back, there is a problem of integrity.



The 6 data quality dimensions with examples

1. Completeness

How complete is your dataset?

2. Accuracy

Is the data an accurate representation of the element it describes?

3. Consistency

Is your data synchronized across your organization?

4. Validity

Does the data fit the defined range and definition?

5. Uniqueness

Are there any duplicates within your data?

6. Integrity

Can your data be traced and connected across the organization?

Data quality goes beyond what cleaning can do...

Let's see what we can clean!

Duplicates

+ Member-only story

Practical Guide to Data Analysis and Preprocessing

Techniques for data cleaning, transformation, and validation to ensure quality data



Leo Anello · Follow

Published in Towards Data Science · 49 min read · Oct 31, 2024

Duplicate

The repetition of identical data points.

Complete case: Rows are identical over all columns

```
# 23. Using the duplicated() method to create a series of boolean values
duplicates = df.duplicated()

# 24. Displaying the duplicated rows
df[duplicates]
```

Find the complete duplicates.
Remove rows so only one datapoint is left.

Near-duplicates are more challenging!



ID	Name	Gender	Salary (\$)	Loan Amount (\$)	Loan Outcome	Loan Type
1	Ms. Phi	F	120,000	48,000	No default	Car
2	Mr. Psi	M	100,000	Unknown	No default	Student loan
3	Ms. Tau	F	(40,000)	16,000	No default	Mortgage
4	Mr. Epsilon	F	90,000	36,000	Defaulted	Car
5	Mr. Rho	M	83,000	33,200	No default	
6	Mr. Rho	M	83,000	33,200	No default	Mortgage
7	Ms. Chi	F	95,000	38,000	No default	Mortgage

Understanding Data Duplicates

Deep dive on duplicates, record hashing, de-duplication, and edge cases



Written by Openbridge Support
Updated over a year ago

Order ID	Customer ID	Product ID		Quantity	Order Date
1	1234	5678		2	2021-01-01
1	1234	5678		2	2021-01-01
1	1234	5678		2	2021-01-01
1	1234	5678		2	2021-01-01
1	1234	5678		2	2021-01-01

Careful with transactional data!

False Positives

Order Item ID	Order ID	Customer ID	Product ID		Quantity	Order Date
1	1	1234	5678		2	2021-01-01
2	1	1234	5678		2	2021-01-01
3	1	1234	5678		2	2021-01-01
4	1	1234	5678		2	2021-01-01
5	1	1234	5678		2	2021-01-01

In the same order, the customer bought 5 items.

Order Item ID	Order ID	Customer ID	Product ID		Quantity	Order Date
1	1	1234	5678		2	2021-01-01
2	1	1234	5678		2	2021-01-01
3	1	1234	5678		2	2021-01-01
4	1	1234	5678		2	2021-01-01
5	1	1234	5678		2 10	2021-01-01

If too fine grain, perhaps transactional data need to be grouped for data analysis

On Jan 1, 2023, there was a product called Red Widget Max in your listings:

Product ID	Product Title	Price	Timestamp
1	Red Widget Max	\$10	2023-01-01

However, on March 1, 2023, a product manager decided to change the Product title to Red Widget Ultimate

Product ID	Product Title	Price	Timestamp
1	Red Widget Ultimate	\$10	2023-03-01

Slowly Changing Data

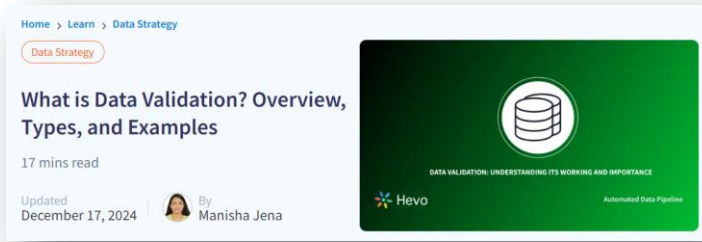
This will result in two rows for the same product ID:

Product ID	Product Title	Price
1	Red Widget Max	\$10
1	Red Widget Ultimate	\$10

Some ID fields should be unique... so how to deal with this?

Duplicates	
What can cause them?	Human entry error Dataset merging Slow changing data
What harm can they do?	Can bias an analysis by inflating counts.
Easy cases	Identical rows (complete cases)
Difficult cases	Near-duplicates (often require domain expertise for plausibility) Duplicates arising from slow changing data
Cleaning errors	False positives (removing non-duplicates) False negatives (not-removing duplicates)

Invalid data



The following are the common Types:

- 1) Data Type Check**
- 2) Code Check**
- 3) Range Check**
- 4) Format Check**
- 5) Consistency Check**
- 6) Uniqueness Check**
- 7) Presence Check**
- 8) Length Check**
- 9) Look Up**

Validation at entry time versus post-processing (cleaning)

Link to UI and validation (front end/Surveys) – if NOT done there... lots of additional work in the back



Data Type

Input validation

- 1) Data Type Check
- 2) Code Check
- 3) Range Check
- 4) Format Check
- 5) Consistency Check
- 6) Uniqueness Check
- 7) Presence Check
- 8) Length Check
- 9) Look Up

A **Data Type check** ensures that data entered into a field is of the correct data type. A field, for example, may only accept numeric data. The system should then reject any data containing other characters, such as letters or special symbols, and an error message should be displayed.



Dataset cleaning

Name	Age	Salary	Join_Date	Is_Manager
Alice	25	50000.0	2020-05-10	True
Bob	Unknown	45000.0	2021-07-15	False
Charlie	30	None	Invalid Date	Yes
Diana	28.5	"60000"	2022-03-20	No
Edward	27	55000.0	2023-01-01	TRUE

Should fit these types



Name	object
Age	float64
Salary	float64
Join_Date	datetime64[ns]
Is_Manager	bool

we need these types



Codes

Input validation

- 1) Data Type Check
- 2) Code Check
- 3) Range Check
- 4) Format Check
- 5) Consistency Check
- 6) Uniqueness Check
- 7) Presence Check
- 8) Length Check
- 9) Look Up

A **Code Check** ensures that a field is chosen from a valid list of values or that certain formatting rules are followed. For example, it is easier to verify the validity of a postal code by comparing it to a list of valid codes. Other items, such as country codes and NAICS industry codes, can be approached in the same way.



Dataset cleaning

Harmonized System

From Wikipedia, the free encyclopedia

The **Harmonized Commodity Description and Coding System (HS)** of **tariff nomenclature** is an internationally standardized system of names and numbers for classifying traded products developed and maintained by the **World Customs Organization (WCO)** (formerly the Customs Co-operation Council), an independent intergovernmental organization with over 170 member countries based in **Brussels, Belgium**.

Example for import/export

01-05 [Animal & Animal Products](#)
 06-15 [Vegetable Products](#)
 16-24 [Foodstuffs](#)
 25-27 [Mineral Products](#)
 28-38 [Chemicals & Allied Industries](#)
 39-40 [Plastics / Rubbers](#)
 41-43 [Raw Hides, Skins, Leather, & Furs](#)
 44-49 [Wood & Wood Products](#)
 50-63 [Textiles](#)
 64-67 [Footwear / Headgear](#)
 68-71 [Stone / Glass](#)
 72-83 [Metals](#)
 84-85 [Machinery / Electrical](#)
 86-89 [Transportation](#)
 90-97 [Miscellaneous](#)
 98-99 [Service](#)

07.10 **Vegetables (uncooked or cooked by steaming or boiling in water), frozen.**

0710.10 - Potatoes

- Leguminous vegetables, shelled or unshelled:

0710.21 -- Peas (*Pisum sativum*)0710.22 -- Beans (*Vigna spp.*, *Phaseolus spp.*)

0710.29 -- Other

0710.30 - Spinach, New Zealand spinach and or ache spinach (garden spinach)

0710.40 - Sweet corn

0710.80 - Other vegetables

0710.90 - Mixtures of vegetables

Example for import/export

22.02  **Waters, including mineral waters and aerated waters, containing added sugar or other sweetening matter or flavoured, and other non-alcoholic beverages, not including fruit or vegetable juices of heading 20.09.**

2202.10 - Waters, including mineral waters and aerated waters, containing added sugar or other sweetening matter or flavoured

2202.90 - Other

22.03  **Beer made from malt.**

2203.00 no subheadings

22.04  **Wine of fresh grapes, including fortified wines; grape must other than that of heading 20.09.**

2204.10 - Sparkling wine

ICD-10 Version:2019

Search

- ▼ ICD-10 Version:2019
 - ▶ I Certain infectious and parasitic diseases
 - ▶ II Neoplasms
 - ▶ III Diseases of the blood and blood-forming organs and certain disorders involving the immune mechanism
 - ▶ IV Endocrine, nutritional and metabolic diseases
 - ▶ V Mental and behavioural disorders
 - ▶ VI Diseases of the nervous system
 - ▶ VII Diseases of the eye and adnexa
 - ▶ VIII Diseases of the ear and mastoid process
 - ▶ IX Diseases of the circulatory system
 - ▶ X Diseases of the respiratory system
 - ▶ XI Diseases of the digestive system
 - ▶ XII Diseases of the skin and subcutaneous tissue
 - ▶ XIII Diseases of the musculoskeletal system and connective tissue
 - ▶ XIV Diseases of the genitourinary system
 - ▶ XV Pregnancy, childbirth and the puerperium
 - ▶ XVI Certain conditions originating in the perinatal period
 - ▶ XVII Congenital malformations, deformations and chromosomal abnormalities
 - ▶ XVIII Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified
 - ▶ XIX Injury, poisoning and certain other consequences of external causes
 - ▶ XX External causes of morbidity and mortality
 - ▶ XXI Factors influencing health status and contact with health services
 - ▶ XXII Codes for special purposes

Example for patient discharge coding (insurance)

- ▼ VII Diseases of the eye and adnexa
 - ▶ H00-H06 Disorders of eyelid, lacrimal system and orbit
 - ▼ H10-H13 Disorders of conjunctiva
 - ▼ H10 Conjunctivitis
 - H10.0 Mucopurulent conjunctivitis
 - H10.1 Acute atopic conjunctivitis
 - H10.2 Other acute conjunctivitis
 - H10.3 Acute conjunctivitis, unspecified
 - H10.4 Chronic conjunctivitis
 - H10.5 Blepharoconjunctivitis
 - H10.8 Other conjunctivitis
 - H10.9 Conjunctivitis, unspecified
 - ▶ H11 Other disorders of conjunctiva
 - ▶ H13 Disorders of conjunctiva in diseases classified elsewhere
 - ▶ H15-H22 Disorders of sclera, cornea, iris and ciliary body



Range

Input validation

- 1) Data Type Check
- 2) Code Check
- 3) Range Check
- 4) Format Check
- 5) Consistency Check
- 6) Uniqueness Check
- 7) Presence Check
- 8) Length Check
- 9) Look Up

A **Range Check** will determine whether the input data falls within a given range. Latitude and longitude, for example, are frequently used in geographic data. Latitude should be between -90 and 90, and longitude should be between -180 and 180. Any values outside of this range are considered invalid.



Dataset cleaning

need to know range

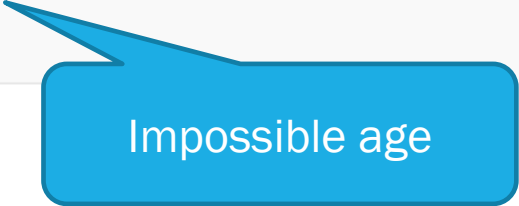


Negative value

1	ID	Name	Gender	Salary (\$)	Loan amount (\$)	Loan Outcome	Loan Type
2	1	Ms. Phi	F	120,000	48,000	No default	Car
3	2	Mr. Psi	M	100,000	Unknown	No default	Student loan
4	3	Ms. Tau	F	(40,000)	16,000	No default	Mortgage
5	4	Mr. Epsilon	F	90,000	36,000	Defaulted	Car
6	5	Mr. Rho	M	83,000	33,200	No default	
7	6	Mr. Rho	M	83,000	33,200	No default	Mortgage
8	7	Ms. Chi	F	95,000	38,000	No default	Mortgage

cannot be
negative


```
# Example in Python: Removing invalid data based on conditions  
df = df[df['age'] > 0]
```



Impossible age

Employee_ID	Age	Salary	Hours_Worked_Per_Week
E001	25	50000	40
E002	16	45000	50
E003	120	60000	38
E004	35	250000	20
E005	30	-1000	70

Out-of-range or outlier?

What are the ranges?



Format

Input validation

- 1) Data Type Check
- 2) Code Check
- 3) Range Check
- 4) Format Check
- 5) Consistency Check
- 6) Uniqueness Check
- 7) Presence Check
- 8) Length Check
- 9) Look Up

Many data types have a predefined format. A Format Check will ensure that the data is in the correct format. Date fields, for example, are stored in a fixed format such as “YYYY-MM-DD” or “DD-MM-YYYY.” If the date is entered in any other format, it will be rejected.



Dataset cleaning

Use of standards

For instance, if a dataset has a “Date” column with both “YYYY-MM-DD” and “MM/DD/YYYY” formats, you can standardize them into a single format for uniformity.

```
# Example in Python: Converting dates to a single format  
df['date'] = pd.to_datetime(df['date'], format='%Y-%m-%d')
```

Non-interpretable text format

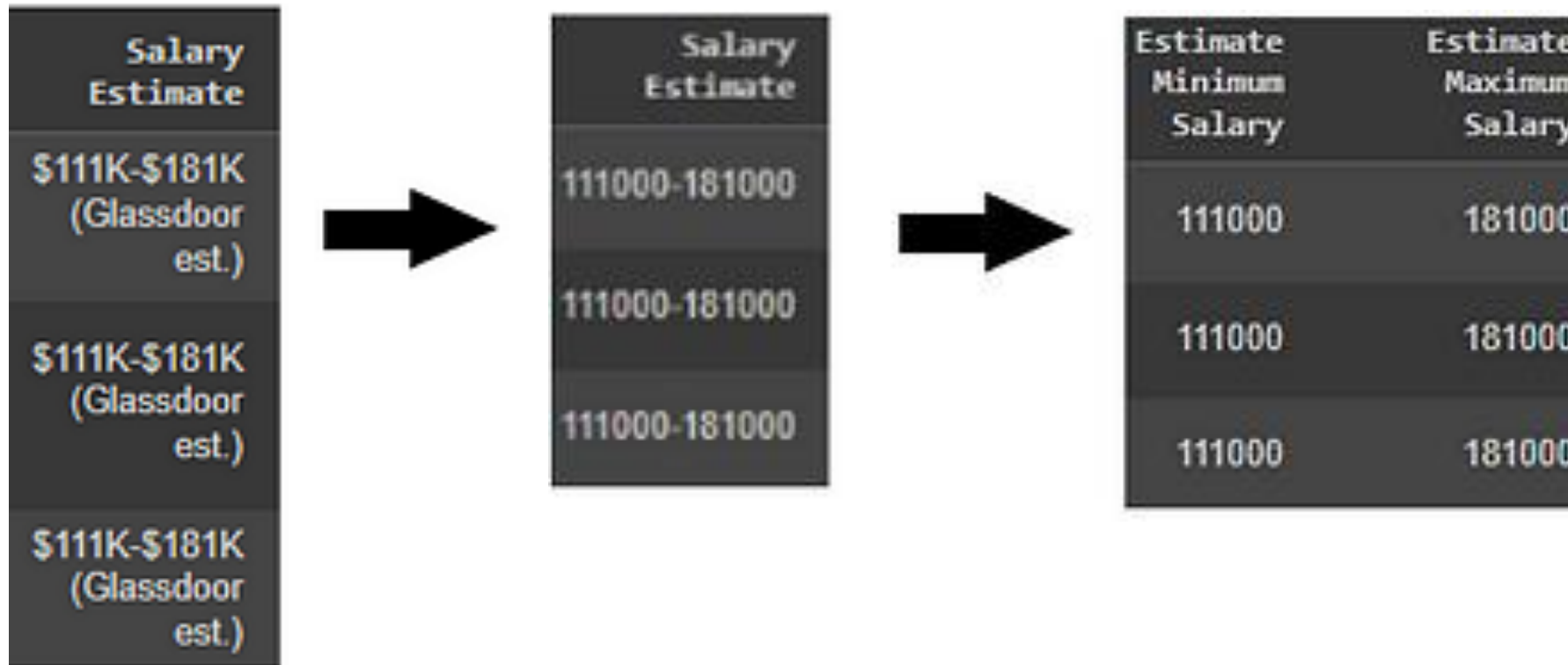
Company Name	Location
Hopper\n3.5	New York, NY
Noom US\n4.5	New York, NY
Sapphire Digital\n3.4	Lyndhurst, NJ



Company Name	Location
Hopper	New York
Noom US	New York
Sapphire Digital	Lyndhurst

Example of a string expressing a numerical range.

Non-interpretable text format





Consistency

Input validation

- 1) Data Type Check
- 2) Code Check
- 3) Range Check
- 4) Format Check
- 5) Consistency Check
- 6) Uniqueness Check
- 7) Presence Check
- 8) Length Check
- 9) Look Up

A **Consistency Check** is a type of logical check that ensures data is entered in a logically consistent manner. Checking if the delivery date for a parcel is after the shipping date is one example.



Business rules

Dataset cleaning



Loan outcome depends on
loan amount

shouldn't be
available

1	ID	Name	Gender	Salary (\$)	Loan Amount (\$)	Loan Outcome	Loan Type
2	1	Ms. Phi	F	120,000	48,000	No default	Car
3	2	Mr. Psi	M	100,000	Unknown	No default	Student loan
4	3	Ms. Tau	F	(40,000)	16,000	No default	Mortgage
5	4	Mr. Epsilon	F	90,000	36,000	Defaulted	Car
6	5	Mr. Rho	M	83,000	33,200	No default	
7	6	Mr. Rho	M	83,000	33,200	No default	Mortgage
8	7	Ms. Chi	F	95,000	38,000	No default	Mortgage

Salary is dependent of
company's rules

Employee ID	Job Title	Salary (\$)
201	Intern	25,000
202	Senior Engineer	120,000
203	CEO	250,000
204	Junior Analyst	180,000

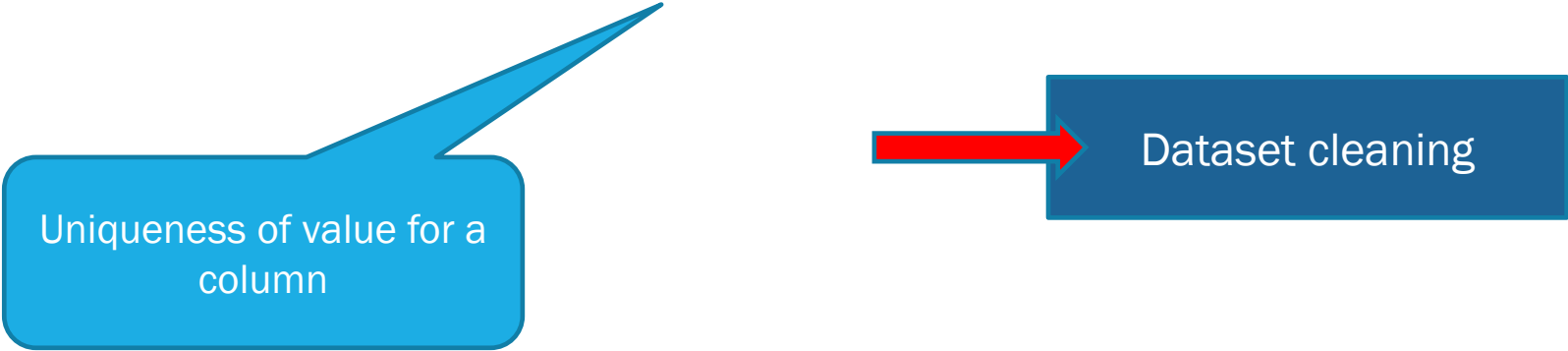


Uniqueness

Input validation

- 1) Data Type Check
- 2) Code Check
- 3) Range Check
- 4) Format Check
- 5) Consistency Check
- 6) Uniqueness Check
- 7) Presence Check
- 8) Length Check
- 9) Look Up

Some data, such as IDs or e-mail addresses, are inherently unique. These fields in a database should most likely have unique entries. A **Uniqueness Check** ensures that an item is not entered into a database more than once.



Uniqueness of value for a column

Dataset cleaning

Employee ID	Name	Email	Job Title	Salary (\$)
401	Emily Rivera	info@travelco.com	Tour Guide	45,000
402	James Carter	james@travelco.com	Travel Agent	55,000
403	Michael Lee	info@travelco.com	Hotel Manager	80,000
404	Sophia Adams	sophia@travelco.com	Event Coordinator	50,000

Are emails really unique? Credit card number? Phone number?



Presence

Input validation

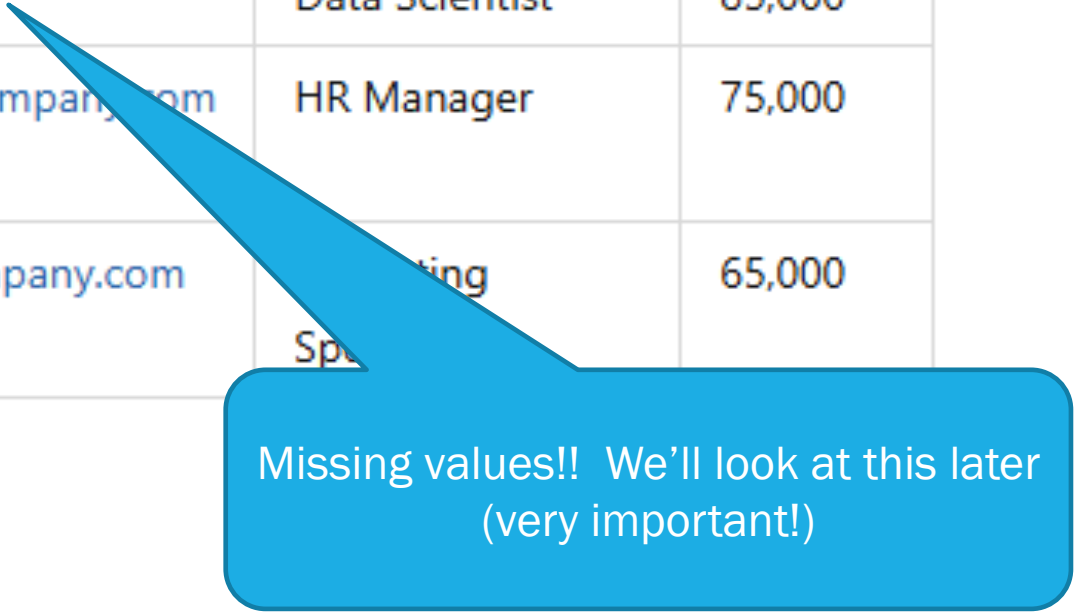
- 1) Data Type Check
- 2) Code Check
- 3) Range Check
- 4) Format Check
- 5) Consistency Check
- 6) Uniqueness Check
- 7) Presence Check
- 8) Length Check
- 9) Look Up

A **Presence Check** ensures that all mandatory fields are not left blank. If someone tries to leave the field blank, an error message will be displayed, and they will be unable to proceed to the next step or save any other data that they have entered. A key field, for example, cannot be left blank in most databases.



Dataset cleaning

Employee ID	Name	Email	Job Title	Salary (\$)
901	Alice Green	alice@company.com	Software Engineer	95,000
902	Bob White		Data Scientist	85,000
903	Charlie Black	charlie@company.com	HR Manager	75,000
904	Dana Blue	dana@company.com	Marketing Specialist	65,000



Missing values!! We'll look at this later
(very important!)



Length



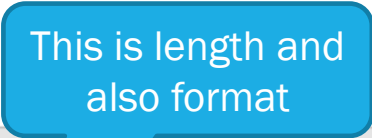
Input validation

- 1) Data Type Check
- 2) Code Check
- 3) Range Check
- 4) Format Check
- 5) Consistency Check
- 6) Uniqueness Check
- 7) Presence Check
- 8) Length Check
- 9) Look Up

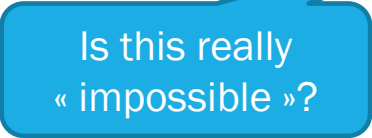
A **Length Check** ensures that the appropriate number of characters are entered into the field. It verifies that the entered character string is neither too short nor too long. Consider a password that must be at least 8 characters long. The Length Check ensures that the field is filled with exactly 8 characters.



Dataset cleaning



Employee ID	Name	Phone Number	Email	Job Title	Salary (\$)
701	John A. Smith	123-45-6789	john.smith@email.com	Software Engineer	85,000
702	Olivia	123-4	olivia@email.com	Data Analyst	75,000
703	Samuel Johnson	987-654-3210	samuel.johnson@email.com	HR Specialist	60,000
704	Catherine Elizabeth Alexandra Montgomery-Smith	555-123-4567	catherine@email.com	Project Manager	95,000



Is this really
« impossible »?



Look-up

Input validation

- 1) Data Type Check
- 2) Code Check
- 3) Range Check
- 4) Format Check
- 5) Consistency Check
- 6) Uniqueness Check
- 7) Presence Check
- 8) Length Check
- 9) Look Up

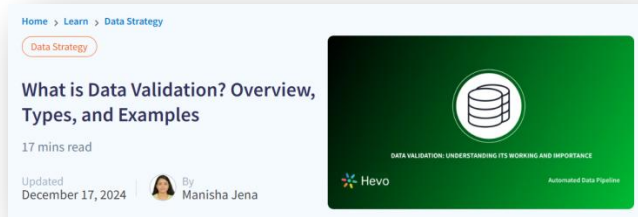
Look Up assists in reducing errors in a field with a limited set of values. It consults a table to find acceptable values. The fact that there are only 7 possible days in a week, for example, ensures that the list of possible values is limited.



Dataset cleaning

Product_ID	Category	Price_Code	Region_Code	Stock_Status
P001	Electronics	A	North_1	In_Stock
P002	Electronics	B	N_1	InStok
P003	Clothing	C	South_1	Out_of_Stock
P004	Clothing	4	S_1	OutofStock
P005	Electronics	B+	East_1	In_Stock

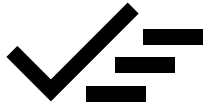
Don't let people type, provide a selection in a menu/list



The following are the common Types:

- 1) Data Type Check
- 2) Code Check
- 3) Range Check
- 4) Format Check
- 5) Consistency Check
- 6) Uniqueness Check
- 7) Presence Check
- 8) Length Check
- 9) Look Up

Validation problems	
What can cause them?	
What harm can they do?	
Easy cases	
Difficult cases	
Cleaning errors	



DATA CLEANING

- Data wrangling
- Data quality
- Data cleaning
 - Duplicates
 - Invalid data
 - Outliers

Part 1

Part 2